TECHNICAL DESIGN & INTERFACE SPECIFICATION

Production Planning Optimizer

Constructive Heuristic + MILP | V6.3.4

DOCUMENT CONTROL	REFERENCE
Version	6.3.4-CHANGEOVER-FIRST-MIN80-NO-GLOBAL-FLOOR
Implementation	Google Colab / Jupyter Notebook (.ipynb)
Authoritative configuration	Excel workbook (.xlsx), 11 required sheets
Solver	SciPy optimize.milp (HiGHS), plus NumPy / optional Numba
Interface	Public GitHub Excel download → planner → four-sheet Excel output
Specification date	12 October 2026

CRITICAL DESIGN RULE Each product has a hard minimum production requirement (80% in the current workbook). There is NO independent global minimum production constraint. Changeovers are optimized first; additional output is optional up to each product target.

1. Purpose, scope, and source baseline

This document describes the behavior implemented in the current V6.3.4 notebook and its matching input workbook, including the input schema, constraint model, search process, solver stages, output workbook, and operational checks. It documents the source implementation rather than proposing an alternative scheduling design.

The planner assigns products and production quantities to eligible machines across a six-day, two-shift-per-day calendar. It schedules one product per machine-shift, enforces production sequencing and capacity rules, accounts for preparation time, and produces a validated Excel plan.

1.1 Current test configuration

Dimension	Current value	Interpretation
Products	10	All ten products appear in targets
Active machines	5	CS1, CS2, EB1, EB2, CS3
Eligible machine/product pairs	21	Out of 50 listed combinations

Dimension	Current value	Interpretation
Planning horizon	6 days / 12 shift positions	Each machine has an explicit calendar row per slot
Total production target	205 t	Upper bound across all products
Sum of individual minimums	164 t	Arithmetic consequence of ten 80% minima; NOT an extra global floor
Preparation time	40 min default	Machine-product value is authoritative

2. End-to-end system architecture

Stage	Implemented processing	Primary artifact
1 / Download	Validate HTTPS GitHub link; convert /blob/ to RAW; download and check XLSX/ZIP signature.	Input XLSX
2 / Parse & validate	Read exactly 11 named sheets; verify headers, keys, references, formulas and rule compatibility.	Typed arrays + constraints
3 / Construct	Generate randomized, distinct candidate plans with Numba acceleration when available.	Feasible candidate archive
4 / MILP-1	Optimize changeovers first, then unmet target tons, then preparation minutes.	Selected schedule
5 / MILP-2	Re-sequence and rebalance on selected machine-product pair sets; accept only improvements that validate.	Refined schedule
6 / Verify & export	Independently check operational constraints, format plan, and write four visible sheets.	<input_stem>_output.xlsx

The input workbook is the single configuration interface. The notebook does not require a JSON input file, a commercial license, or a separate application server. A public GitHub repository is used only to retrieve the XLSX file; the optimization runs in the Python kernel.

3. Optimization model and objective hierarchy

3.1 Index sets and principal variables

Symbol	Meaning
m / p / t	Machine / product / calendar slot (day, shift).
x[m,p,t]	Binary indicator: product p is active on machine m in slot t.
q[m,p,t]	Continuous planned production quantity in metric tons.
y[m,p]	Binary indicator: product p is used on machine m during the horizon.
start[m,p,t]	Binary indicator: the product run starts in the slot and incurs setup.

Symbol	Meaning
change[m,t]	Binary indicator: a genuine product transition on that machine.
unmet[p]	Continuous shortfall relative to product p target; limited by individual minimum.
exception[m,p]	Binary indicator for the narrowly permitted final below-minimum run.

3.2 Hard constraints

Constraint	Operational interpretation / representative condition
Per-product minimum	Sum of scheduled tons for each p ≥ Target_t[p] × MinRequired_pct[p]. Current rate: 0.80.
Per-product upper bound	Sum of scheduled tons for each p ≤ Target_t[p]; surplus production is disabled.
Slot exclusivity	For each machine and shift, at most one product is active.
Time balance	Production_t / Capacity_tph + Preparation_min / 60 ≤ calendar.Hours.
Eligibility	A machine-product assignment is permitted only when Allowed=1 and the rate is positive.
Daily product tonnage	Quantity produced for a given machine/product/day ≤ MaxDaily_t.
Batch minimum	A used machine-product run must satisfy MinRun_t, allowing the documented final remainder exception.
Sequence	Explicit same-machine predecessor relationships must occur in the specified order.
Daily transitions	At most Max_Changeovers_Per_Day transitions per machine-day; current value is 1.
No-return	A machine cannot produce A, switch to B, then return to A during the horizon.
Initial state	Carried product/setup state affects initial setup and first product transition.
Hard targets / allocation	TargetMode=HARD requires the full target. Allocation splits are hard only if configured as such.

NO GLOBAL TONNAGE GUARD MinimumPlanned_t is absent from the current input schema and the current code. The sum of product minima is 164 t for this dataset, but that total is derived from ten separate hard constraints. The optimizer is not allowed to compensate for one underproduced product by overproducing another.

3.3 Lexicographic priorities

Priority	Criterion	Behavior
1	MIN_CHANGEOVERS	Minimize product-to-product transitions; an initial empty-machine setup is not a changeover.
2	MIN_UNMET	At the selected transition count, minimize unmet tons (maximize production toward targets).

Priority	Criterion	Behavior
3	MIN_TOTAL_PREPARATION	Reduce setup minutes without sacrificing the first two goals.
4	MIN_PRODUCT_MACHINE_SPLITS	Soft ranking preference: favor fewer machines per product.
5	MIN_FAMILY_MACHINE_SPLITS	Soft ranking preference: group family production when possible.
6	MIN_MACHINE_PRIORITY_COST	Final tie-break preference based on machine-product preference scores.

Implementation nuance: the primary MILP explicitly solves the first three objectives in separate stages with incumbent bounds. Product/family splits and machine preference appear in candidate ranking and the selected-plan comparison; they are not all solved as separate, globally certified MILP objectives in this notebook.

4. Computational procedure

4.1 Constructive heuristic

A randomized constructive routine assembles feasible production blocks while considering eligible machines, planning slots, minimum run sizes, target balances, predecessor rules and machine-product capacities. When CriticalProductFirst=1, machines with fewer eligible choices receive heuristic emphasis. Numba JIT acceleration is used if available.

Heuristic property	Current implementation
Unique trial target	HeuristicTrials = 600,000 (configured in settings).
Deduplication	Fingerprint of machine/slot assignment and quantities rounded to 0.01 t.
Raw draw limit	Up to four times the requested unique-plan count.
Feasibility gate	Each product must independently meet its minimum; candidate then passes the same independent validation.
Archive	Best valid candidate per distinct machine-product allocation topology; up to 300 retained.
Seed	RandomSeed = 20261008 in the supplied workbook; seed 0 requests a fresh random seed.

If the heuristic archive has no feasible plan, MILP-1 attempts a feasibility search. The code does not silently lower a product minimum to make the model feasible.

4.2 MILP-1: free machine allocation and re-sequencing

The primary MILP is constructed with SciPy optimize.milp and solved by HiGHS. Machine-product assignments, shift positions, product order, and continuous tons are optimization variables. No additional global transition-count cap or total-tonnage minimum is introduced.

Substage	Goal / safeguard
Feasibility fallback	If no heuristic incumbent exists, solve an unmet-target model to obtain a valid schedule.
Changeovers	Minimize product changes, constrained not to exceed the known incumbent transition count.
Throughput	Minimize unmet tonnage while bounding the transition count to the best selected level.
Preparation	Optionally minimize setup time while bounding both transitions and unmet tons.
Validation	Decode MILP variables to a schedule; reject candidates that fail independent checks.

4.3 MILP-2, normalization and final acceptance

MILP-2 freezes the machine-product pairs of promising incumbent allocations, then re-optimizes shift order and quantities. Its target is minimum unmet tons; a proposal replaces the current schedule only if it passes the independent validator and improves the configured lexicographic ranking. A final normalization step can move uninterrupted single-product operation into earlier shifts without changing output or changeover counts.

Solver time parameters are maximum budgets, not guaranteed elapsed durations. Solver status, MIP gap and available bounds are logged, but a plan must not be described as mathematically proven optimal unless the relevant solver stage reports an appropriate optimality certificate.

5. Input workbook interface (11 required sheets)

The parser requires the precise sheet set shown below and exactly the specified header names (column order is flexible). Unknown sheets, missing headers, unsupported formulas, duplicate keys and unsupported settings are treated as input errors. The workbook format is a normal, unfiltered XLSX with data in row 1 onward.

5.1 products

Product master data and salt/sequence metadata.

Field	Meaning / validation rule
Product_ID	Unique product key, for example P001.
Product	Product display name.
Family	Family used for sequence and grouping.
Salt_Status	UNSALTED, SALTED or NONE.
Sequence_Order	Positive same-family order number.
Priority	Product priority tier; all supplied values must be equal.

5.2 machines

Machine master and activation settings.

Field	Meaning / validation rule
Machine_ID	Unique machine key.
Machine	Machine display name (e.g., CS1).
Active	1=eligible for planning; 0=excluded.
Priority	Machine-level preference score (positive integer).

5.3 machine_product

Eligibility and process parameters for each machine-product pair.

Field	Meaning / validation rule
Machine_ID	Reference to machines.Machine_ID.
Product_ID	Reference to products.Product_ID.
Allowed	1=permitted; 0=forbidden.
MachinePriority	Positive preferred-machine score when Allowed=1.
Capacity_tph	Nominal tons per production hour.
MinRun_t	Minimum run quantity (t), except narrow final-run rule.
MaxDaily_t	Maximum tonnage for this pair on one day.
Preparation_min	Setup minutes at the start of a new product run.

FORMULA CONVENTION The provided sheet uses =E<row>*3 for MinRun_t and =E<row>*8 for MaxDaily_t. The parser also supports the original same-row structured reference format. Unsupported formulas are rejected, rather than relying on potentially stale caches.

5.4 targets

Demand/target and product-specific compulsory percentage.

Field	Meaning / validation rule
Product_ID	One row per product.
Product	Must match products.Product exactly.
Target_t	Target quantity in tons.
TargetPriority	Positive tier; mixed tiers not implemented.
TargetMode	SOFT or HARD. HARD requires full target.
MinRequired_pct	Decimal fraction between 0 and 1; current value 0.80.

Field	Meaning / validation rule
MinRequired_t	Calculated minimum (Target_t × MinRequired_pct).

FORMULA CONVENTION MinRequired_t is written as =C<row>*F<row> and recomputed by the Python parser. The current ten MinRequired_pct values are all 0.80. SOFT means the remaining 20% can remain unmet; it does not relax the 80% hard minimum.

5.5 calendar

Explicit availability for every active machine/slot.

Field	Meaning / validation rule
Machine_ID	Machine key.
Day	Day-1, Day-2, …; sorted numerically.
Shift	Positive integer shift identifier.
Available	1=open, 0=closed.
Hours	Usable hours. Closed slots must have Hours=0.
Reason	Human-readable calendar explanation.

5.6 sequence_rules

Explicit hard within-machine predecessor pairs.

Field	Meaning / validation rule
Rule_ID	Unique rule identifier.
Family	Product family.
Before_Product_ID	Product that must come first.
After_Product_ID	Product that must come later.
Scope	SAME_MACHINE (supported).
Hard	Must be 1 in this version.

5.7 changeover_rules

Machine-day transition and single-product-per-shift policy.

Field	Meaning / validation rule
Rule_ID	Unique rule identifier.

Field	Meaning / validation rule
Scope	MACHINE_DAY; one global rule in this version.
Max_Products_Per_Shift	Must be 1.
Max_Changeovers_Per_Day	Allowed changeovers per machine-day (current: 1).
Hard	Must be 1.
Description	Rule documentation; not executable code.

5.8 allocation_rules

Preferences or optional hard limits for product/family splitting.

Field	Meaning / validation rule
Rule_ID	Unique rule identifier.
Scope	ALL_PRODUCTS, PRODUCT:<ID>, ALL_FAMILIES or FAMILY:<name>.
Prefer_Single_Machine	One-machine preference (1 in the current contract).
Preferred_Max_Machines	Preferred value 1 in current implementation.
Allow_Split_If_Required	1 allows splitting as needed.
Hard	1 would make the one-machine limit a hard rule.
Description	Explanatory text.
Objective_ID	Corresponding product- or family-split objective.

5.9 initial_state

Machine state at the start of the planning horizon.

Field	Meaning / validation rule
Machine_ID	One row for every machine, active or inactive.
Current_Product_ID	Previously running product; blank if none.
Run_Produced_t	Already-produced quantity credited to current run.
Setup_Completed	1 means initial setup for that same product is complete.

5.10 objectives

Enabled objective IDs in lexicographic order.

Field	Meaning / validation rule
Priority	Ascending objective order.
Objective_ID	Implemented objective name.
Direction	MIN only.
Enabled	1=enabled, 0=disabled.
Description	Readable purpose of the objective.

5.11 settings

Typed runtime and validation configuration.

Field	Meaning / validation rule
Parameter	Exact recognized configuration name.
Value	Numeric, flag, or supported enum value.
Description	Human-readable explanation.

6. Settings reference

All 28 settings rows are part of the strict input contract; changing or removing an unsupported setting may cause validation to stop. The descriptions below represent their implemented effect in this version, including legacy fields that do not impose constraints.

Parameter	Current value	Meaning
SchemaVersion	6.0	Input schema identifier; must be 6.0.
ShiftHours	4	Nominal shift duration; actual per-slot hours come from calendar.Hours.
DefaultPreparationMin	40	Fallback setup minutes when a pair-specific Preparation_min is blank.
MaxPlanningDays	6	Upper bound for parsed Day-n calendar labels.
AllowFinalRunBelowMinimum	1	Permit a strictly controlled final below-minimum run that completes the product target.
AllowSurplusProduction	0	Must be 0; above-target production is forbidden.
PreventProductReturn	1	Must be 1; A→B→A prohibited even across idle/closed slots.
PreferContinuousProduction	1	Tie-break preference to reduce gaps within production.
CriticalProductFirst	1	Constructive heuristic emphasis for restricted products/machines.

Parameter	Current value	Meaning
ObjectiveMode	LEXICOGRAPHIC_CHANGEOVER_FIRST	Must be LEXICOGRAPHIC_CHANGEOVER_FIRST.
UnmetTolerance_t	0	Legacy dashboard parameter; not a global unmet/production constraint.
UnmetToleranceScope	TOTAL_PLAN	Legacy compatibility value; must be TOTAL_PLAN.
UnmetReferenceMode	BEST_FOUND_BASELINE	Legacy compatibility value; must be BEST_FOUND_BASELINE.
CountCrossDayChangeovers	1	Must be 1; transitions across days are counted.
ChangeoverAttribution	NEW_PRODUCT_START_DAY	Must be NEW_PRODUCT_START_DAY.
CountInitialSetupAsChangeover	0	Must be 0; first setup on empty machine not a product transition.
IdleResetsProductState	0	Must be 0; idle/closed calendar does not reset the product state.
SequenceRuleSource	sequence_rules	Must be sequence_rules.
SequenceConflictPolicy	ERROR	Must be ERROR.
OptimizationMode	HYBRID	Must be HYBRID.
HeuristicTrials	600000	Requested number of distinct constructive candidates.
MILPPolishTimeSec	90	MILP-1 time budget in seconds.
MILPNeighborhoodMode	REALLOCATION_AND_RESEQUENCING	Must be REALLOCATION_AND_RESEQUENCING.
RuntimeLogging	1	Must be 1 for audited logging.
InputValidation	STRICT	Must be STRICT.
UnknownParameterPolicy	ERROR	Must be ERROR.
RandomSeed	20261008	Nonnegative random seed (0 generates a new random seed).
SecondMILPTimeSec	165	MILP-2 time budget in seconds.

7. Output workbook interface (4 visible sheets)

The result filename is derived from the downloaded input workbook: production_planning_input_v6_3_changeover_first_output.xlsx. The exporter temporarily builds an internal Details/Detaylar calculation sheet, then materializes cached output values, removes that sheet, and saves only the four visible sheets described here.

7.1 dashboard

Metric	Meaning
Total_Target_t	Sum of product targets (t).
Planned_t	Total production of the selected schedule (t).
Unmet_t	Total shortfall against targets (t).
Fulfillment	Planned tons / target tons (fraction displayed as percentage).
Runs	Count of machine-product run blocks in the selected schedule.
Changeovers	Count of genuine product-to-product machine transitions.
UnmetTolerance_t	Legacy value from settings; not a hard production threshold.
Heuristic_Attempts	Requested distinct heuristic trial count in the run.
MILP_Budget_sec	Sum of primary and secondary MILP time budgets.

7.2 results-plan

Grid grain: one row per Day and Shift. Each machine contributes four repeated columns: Prep_min, Product, Planned_t, and Run_ID. A product name indicates an active production slot; IDLE indicates unused available time and CLOSED indicates zero calendar availability.

Column group	Interpretation
Day / Shift	Calendar sequence (for example Day-2, shift 1).
Machine name	Displayed above the four columns for that machine.
Prep_min	Setup minutes allocated to this slot; continuation of the same product normally has 0.
Product	Scheduled product name or IDLE / CLOSED.
Planned_t	Tonnes scheduled for that machine/shift.
Run_ID	Human-readable identifier that persists over a continuous same-product run.

7.3 product-summary

Field	Meaning
Product_ID	Product key.
Product	Product display name.
Family	Product family.
TargetMode	SOFT or HARD target policy.
Target_t	Target quantity in tons.

Field	Meaning
Planned_t	Scheduled product quantity in tons.
Unmet_t	Unmet target quantity.
Surplus_t	Amount above target (normally 0).
Fulfillment	Planned / target ratio.
Machines	Comma-separated machine names used.
Machine_Count	Number of machines used for the product.
Runs	Number of machine-product runs.
Target_Status	MET, SOFT_UNMET or equivalent target status.
MinRequired_pct	Product-specific required fraction.
MinRequired_t	Product-specific required tonnage.
Minimum_Status	PASS or FAIL against the hard product minimum.

7.4 machine-summary

Field	Meaning
Machine_ID	Machine key.
Machine	Machine name.
Planned_t	Total planned tonnes on the machine.
Available_h	Total calendar available time.
Production_h	Sum of tonnage / production rate.
Preparation_h	Total setup minutes divided by 60.
Occupied_h	Production hours plus preparation hours.
Idle_h	Available hours minus occupied hours.
Utilization	Occupied_h / Available_h.
Products	Number of distinct products on the machine.
Runs	Number of machine-product runs.
Changeovers	Machine total of genuine product changes.
Max_Daily_Changes	Largest number of changes observed on any day.

Presentation contract: the workbook is intentionally monochrome with black thin borders, ordinary cell ranges rather than filter tables, compact header spacing, no frozen panes, and hidden scrollbars. These choices affect presentation only; they do not change the production mathematics.

8. Execution, dependencies, and failure handling

8.1 Google Colab / Jupyter launch

1. Open the released .ipynb file in Google Colab (or a compatible Jupyter kernel).

2. Run the dependency installation cell. The notebook requests numpy>=1.24, scipy>=1.11, openpyxl>=3.1 and numba>=0.58.

3. Set INPUT_GITHUB_URL to a public GitHub /blob/ XLSX path or raw.githubusercontent.com XLSX URL; the supplied notebook already contains the configured repository path.

4. Run all cells. The notebook downloads the source file into /content (Colab) or the current local directory, replacing the working input copy.

5. The planner validates input, runs the heuristic and MILP stages, and saves <input_stem>_output.xlsx. Colab triggers a browser download; local Jupyter leaves the file in the working directory.

Runtime component	Role
Python 3 / notebook	Execution environment and input parameter cell.
NumPy	Dense arrays, numerical utilities, plan representation.
Numba (optional at import)	JIT compilation of constructive candidate generation; Python fallback exists.
SciPy + HiGHS	Mixed-integer linear programming solver with per-stage time limits.
openpyxl	Input workbook reading, output creation and final formatting.
urllib / zipfile	GitHub HTTPS download and XLSX container sanity check.

8.2 Validation and error response

The input parser rejects an unexpected sheet set, duplicate keys, inconsistent machine/product references, invalid numeric flags, unsupported source formulas, invalid calendar hours, absent sequence dependencies and unsupported objectives/settings. Before export, an independent validator recomputes slot times, daily tonnage, per-product minima, hard targets, per-run minima, sequencing, split limits and transition counts from the decoded plan.

If the minimum production requirements cannot be met, the planner raises an input/feasibility error rather than weakening them. Download failures raise a descriptive exception. The runner will not report successful completion or download a result workbook if main() returns a nonzero exit code.

9. Worked example from the supplied data

For product P001, Target_t = 40.00 t and MinRequired_pct = 0.80; therefore MinRequired_t = 32.00 t. The model must plan at least 32.00 t of P001, but may plan up to 40.00 t if this does not worsen the higher-priority changeover objective. The same independent rule applies to each of the other nine products.

Example measure	Value	Explanation
Total target	205.00 t	Sum of all ten Target_t entries.

Example measure	Value	Explanation
Sum of product minima	164.00 t	205.00 × 80%; descriptive sum, not a separate MILP constraint.
Illustrative test output	195.47 t	From the included short smoke-test output workbook.
Illustrative changeovers	6	From that specific test execution.
Illustrative fulfillment	95.35%	195.47 / 205 rounded for presentation.
Individual minimum statuses	10 PASS	All ten products meet their own minimums in the sample output.

EVIDENCE BOUNDARY These example KPIs come from a short validation run of the current source, not from a fresh full 600,000-candidate benchmark. Results may vary with input data, seed, solver/runtime versions and time budgets. An OPTIMAL solver status is needed before claiming a new global optimum.

10. Known assumptions and current implementation limits

- The sheet set and all recognized headers/settings are strict. The schema identifier is 6.0 even though the implementation release is V6.3.4.
- The supplied target and product priority tiers are uniform; mixed priority tiers are rejected in this code rather than silently reweighted.
- Sequence rules support explicit hard SAME_MACHINE predecessor pairs; same-family sequence metadata is checked for consistency.
- The model prohibits surplus production and return to a previously used product on the same machine, including across idle and closed slots.
- Each shift has at most one product; an unrestricted within-shift changeover model is not implemented.
- A below-minimum final run is permitted only under the implemented completed-target exception; a low available capacity is not, by itself, an exception.
- MILP-2 restricts re-sequencing to selected machine-product allocation sets; it is not a second, independent global search over all assignments.
- Product/family grouping and machine-preference objectives are soft criteria; the code should not be represented as mathematically proving their global optima.
- Use the solver stage status, bound and gap where logged to distinguish a feasible schedule from an optimality certificate.
- Output workbook cells contain frozen values after the intermediate Details sheet is removed. Editing the resulting report will not rerun the optimizer.

11. Traceability and release checklist

Source item	Canonical reference
Implementation notebook	production_planner_v6_3_min80_no_global_floor_colab.ipynb
Input workbook	production_planning_input_v6_3_changeover_first.xlsx

Source item	Canonical reference
GitHub input URL	https://github.com/hadokan/Production-planning/blob/main/production_planning_input_v6_3_changeover_first.xlsx
Notebook SHA-256	d3842b601dfc66ae7f50b8241e89e6dd9d3665aabb3ea69b161f029810cca94c
Input SHA-256	8bead38c46c980d364354b7079057197caf387f9589759b91d177e967ac64ebb
Expected output	production_planning_input_v6_3_changeover_first_output.xlsx

Release check: verify GitHub contains the matching current input workbook, that MinRequired_pct values are explicitly set for all products, that initial_state rows cover all machines, and that the selected output reports Minimum_Status=PASS for every product. Review solver logs when an optimum guarantee is required.

Appendix A. Acceptance and regression tests

The following source-aligned tests are useful when changing the input template or planner code. The table states the expected class of behavior; it is not a claim that this document executed every scenario.

Test condition	Expected planner behavior
Raise one MinRequired_pct above 1.00	Reject target percentage as out of range.
Remove the initial_state worksheet	Reject workbook because the 11-sheet schema is incomplete.
Set calendar Available=0 with Hours>0	Reject conflicting availability and time values.
Change a formula to a different-row capacity reference	Reject unsupported or mismatched formula; do not use a stale Excel cache.
Assign a product to Allowed=0 pair	Reject decoded plan during independent eligibility validation.
Schedule a machine A → B → A	Reject as a prohibited product return, even across idle slots.
Complete normal run with 0-minute same-product continuation	Charge preparation only at startup / product transition.
Build output successfully	Verify four expected sheets and all product minima marked PASS.

Appendix B. Minimal execution logic

Step	Effective program operation
01	Download source XLSX; reject invalid URL or non-XLSX response.
02	Parse strictly; compute each required[p] = target[p] × minimum_pct[p].
03	Generate candidate plans and retain valid distinct topologies.
04	Build and run MILP-1: transitions → unmet tons → setup minutes.
05	Try MILP-2, accept ranked improvements, normalize, validate and export the final Excel.